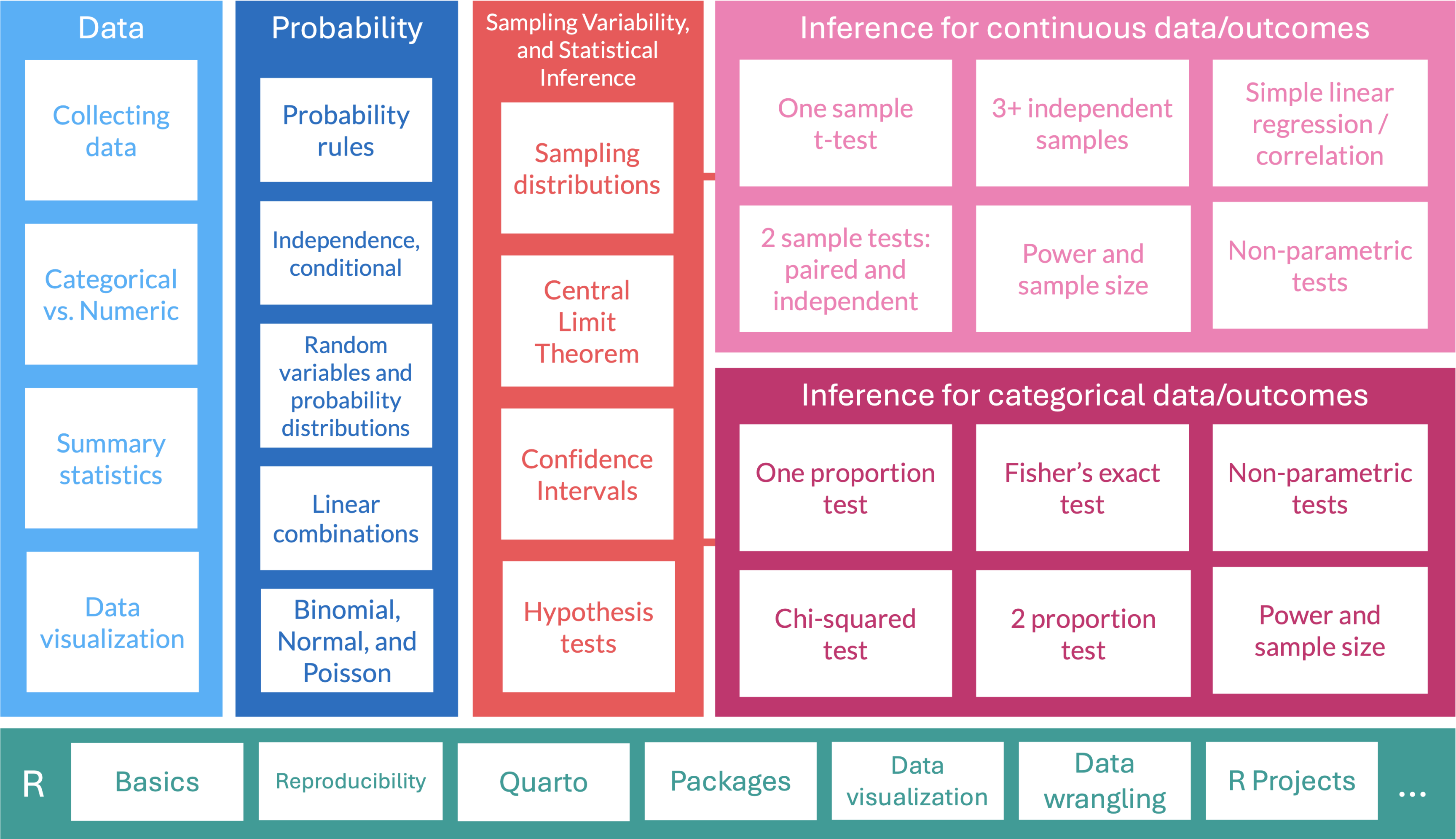


R09: Summarizing data with tidyverse

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Where are we?



What is the tidyverse? (revisited)

The **tidyverse** is a collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

- **ggplot2** - data visualisation
- **dplyr** - data manipulation
- **tidyr** - tidy data
- **readr** - read rectangular data
- **purrr** - functional programming
- **tibble** - modern data frames
- **stringr** - string manipulation
- **forcats** - factors
- and many more ...



Tidy data¹

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	213766	1280425583

variables

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	213766	1280425583

observations

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	213766	1280425583

values

- 1. Each variable must have its own column.
- 2. Each observation must have its own row.
- 3. Each value must have its own cell.

Pipe operator (**magrittr**)

- The pipe operator (**%>%**) allows us to step through sequential functions in the same way we follow if-then statements or steps from instructions

I want to find my keys, then start my car, then drive to work, then park my car.

Nested

```
1 park(drive(start_car(find("keys")),  
2      to = "work"))
```

Piped

```
1 find("keys") %>%  
2   start_car() %>%  
3   drive(to = "work") %>%  
4   park()
```

Using `summarize()`

group_by(): group by one or more variables

- What if I want to quickly look at group differences?
- It will not change how the data look, but changes the actions of following functions

I want to group my data by sex assigned at birth.

```
1 dds.discr5 = dds.discr2 %>%  
2   group_by(SAB)  
3   glimpse(dds.discr5)
```

Rows: 1,000

Columns: 7

Groups: SAB [2]

```
$ id      <int> 10210, 10409, 10486, 10538, 10568, 10690, 10711, 10778, 1...  
$ age.cohort <fct> 13-17, 22-50, 0-5, 18-21, 13-17, 13-17, 13-17, 13-17, 13-...  
$ age      <int> 17, 37, 3, 19, 13, 15, 13, 17, 14, 13, 13, 14, 15, 17, 20...  
$ SAB      <fct> Female, Male, Male, Female, Male, Female, Female, Male, F...  
$ expenditures <int> 2113, 41924, 1454, 6400, 4412, 4566, 3915, 3873, 5021, 28...  
$ R_E      <fct> White not Hispanic, White not Hispanic, Hispanic, Hispani...  
$ exp_to_age <dbl> 124.2941, 1133.0811, 484.6667, 336.8421, 339.3846, 304.40...
```


summarize(): summarize your data or grouped data into one row

- What if I want to calculate specific descriptive statistics for my variables?
- This function is often best used with `group_by()`
- If only presenting the summaries, functions like `tbl_summary()` is better
- `summarize()` creates a new data frame, which means you can plot and manipulate the summarized data

Over whole sample:

```
1 dds.discr2 %>%
2   summarize(
3     ave = mean(expenditures),
4     SD = sd(expenditures),
5     med = median(expenditures))
```

A tibble: 1 × 3

	ave	SD	med
	<dbl>	<dbl>	<dbl>
1	18066.	19543.	7026

Grouped by sex assigned at birth:

```
1 dds.discr2 %>%
2   group_by(SAB) %>%
3   summarize(
4     ave = mean(expenditures),
5     SD = sd(expenditures),
6     med = median(expenditures))
```

A tibble: 2 × 4

	SAB	ave	SD	med
	<fct>	<dbl>	<dbl>	<int>
1	Female	18130.	20020.	6400
2	Male	18001.	19068.	7219

Using `get_summary_stats()`

get_summary_stats() from rstatix package

```
1 dds.discr2 %>% get_summary_stats()
```

```
# A tibble: 4 × 13
```

	variable	n	min	max	median	q1	q3	iqr	mad	mean	sd
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	id	1000	1.02e4	99898	55384.	31809.	76135.	44326	3.27e4	54663.	25644.
2	age	1000	0	95	18	12	26	14	1.04e1	22.8	18.5
3	expendi...	1000	2.22e2	75098	7026	2899.	37713.	34814	7.76e3	18066.	19543.
4	exp_to_...	1000	2.76e1	Inf	462.	274.	938.	664.	3.54e2	Inf	NaN

```
# i 2 more variables: se <dbl>, ci <dbl>
```

```
1 dds.discr2 %>%
```

```
2   group_by(R_E) %>%
```

```
3   get_summary_stats(expenditures, type = "common")
```

```
# A tibble: 8 × 11
```

	R_E	variable	n	min	max	median	iqr	mean	sd	se	ci
	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	American...	expendi...	4	3726	58392	41818.	34085.	36438.	25694.	12847.	40885.
2	Asian	expendi...	129	374	75098	9369	30892	18392.	19209.	1691.	3346.
3	Black	expendi...	59	240	60808	8687	37987	20885.	20549.	2675.	5355.
4	Hispanic	expendi...	376	222	65581	3952	7961.	11066.	15630.	806.	1585.
5	Multi Ra...	expendi...	26	669	38619	2622	2060.	4457.	7332.	1438.	2962.

6	Native H...	expendi...	3	37479	50141	40727	6331	42782.	6576.	3797.	16337.
7	Other	expendi...	2	2018	4615	3316.	1298.	3316.	1836.	1298.	16499.
8	White no...	expendi...	401	340	68890	15718	39157	24698.	20604.	1029.	2023.

How to force all output to be shown? (1/2)

Use `kable()` from the `knitr` package.

```
1 dds.discr2 %>% get_summary_stats() %>% kable()
```

variable	n	min	max	median	q1	q3	iqr	mad
id	1000	10210.000	99898	55384.500	31808.750	76134.750	44326.000	32734.325
age	1000	0.000	95	18.000	12.000	26.000	14.000	10.378
expenditures	1000	222.000	75098	7026.000	2898.750	37712.750	34814.000	7760.670
exp_to_age	1000	27.571	Inf	461.752	273.881	938.125	664.244	353.971

How to force all output to be shown? `knitr` (2/2)

Use `kable()` from the `knitr` package.

```
1 dds.discr2 %>%  
2   group_by(R_E) %>%  
3   get_summary_stats(expenditures, type = "common") %>%  
4   kable()
```

R_E	variable	n	min	max	median	iqr	mean	sd	s
American Indian	expenditures	4	3726	58392	41817.5	34085.25	36438.250	25693.912	12846.95
Asian	expenditures	129	374	75098	9369.0	30892.00	18392.372	19209.225	1691.27
Black	expenditures	59	240	60808	8687.0	37987.00	20884.593	20549.274	2675.28
Hispanic	expenditures	376	222	65581	3952.0	7961.25	11065.569	15629.847	806.04
Multi Race	expenditures	26	669	38619	2622.0	2059.75	4456.731	7332.135	1437.95
Native Hawaiian	expenditures	3	37479	50141	40727.0	6331.00	42782.333	6576.462	3796.92
Other	expenditures	2	2018	4615	3316.5	1298.50	3316.500	1836.356	1298.50

R_E	variable	n	min	max	median	iqr	mean	sd	s
White not Hispanic	expenditures	401	340	68890	15718.0	39157.00	24697.549	20604.376	1028.93



Making a Table 1

Table 1 example

- Often, research studies will show a table with all the summary statistics (lovingly called “Table 1”)
- Basic Table 1 will show all variables with:
 - Mean and SD for the numeric variables
 - $n(\%)$ for categorical variables

Are We on the Same Page?: A Cross-Sectional Study of Patient-Clinician Goal Concordance in Rheumatoid Arthritis

J Barton et al.

Arthritis Care & Research.

2021 Sep 27 <https://pubmed.ncbi.nlm.nih.gov/34569172/>

Table 1. Patient characteristics, overall and by concordance

		Total N=204	Discordant N=40	Concordant N=164	p-value
Site, n (%)	OHSU	122 (62.7%)	26 (65.0%)	96 (62.2%)	0.86
	VA	76 (37.3%)	14 (35.0%)	62 (37.8%)	
Gender, n (%)	Male	85 (41.7%)	18 (45.0%)	67 (40.9%)	0.72
	Female	119 (58.3%)	22 (55.0%)	97 (59.1%)	
Age (years), mean (SD)		57.2 (14.2)	58.2 (15.1)	56.9 (14.0)	0.62
Language, n (%)	English	168 (84.4%)	35 (92.1%)	133 (82.6%)	0.21
	Spanish	31 (15.6%)	3 (7.9%)	28 (17.4%)	
Limited English language proficiency, n (%)		30 (15.1%)	3 (7.9%)	27 (16.8%)	0.17
Coupled, n (%)		110 (57.9%)	22 (61.1%)	88 (57.1%)	0.71
Education, n (%)	High school or less	60 (31.6%)	15 (40.5%)	45 (29.4%)	0.24
	Some college or more	130 (68.4%)	22 (59.5%)	108 (70.6%)	
Income, >\$40,000, n (%)	Less than \$40,000	85 (45.5%)	12 (33.3%)	73 (48.3%)	0.14
	Greater than \$40,000	102 (54.5%)	24 (66.7%)	78 (51.7%)	
People in household, median (IQR)		2 (2-4)	2 (2-3)	2 (2-4)	0.92
Race/Ethnicity, n (%)	White	123 (68.3%)	25 (78.1%)	98 (66.2%)	0.62
	Black	6 (3.3%)	0 (0.0%)	6 (4.1%)	
	Latinx/Hispanic	39 (21.7%)	6 (18.8%)	33 (22.3%)	
	Other	12 (6.7%)	1 (3.1%)	11 (7.4%)	
Limited health literacy, n (%)		55 (28.6%)	13 (35.1%)	42 (27.1%)	0.42
Disease duration (years), median (IQR)		8 (4-16)	13 (5-21)	7 (4-15)	0.039
Number of medications, median (IQR)		1 (1-2)	1 (0-2)	1 (1-2)	0.10
Depressive symptoms, n (%)		38 (20.8%)	3 (8.1%)	35 (24.0%)	0.040
PTSD, n (%)		13 (7.1%)	2 (5.6%)	11 (7.5%)	1.00
Self-efficacy score, mean (SD)		6.3 (2.1)	6.3 (2.1)	6.3 (2.1)	0.96
Trust in Physician, n (%)		106 (53.8%)	19 (51.4%)	87 (%)	0.74
Disease activity score (CDAI), mean (SD)		12.8 (10.5)	10.5 (9.7)	13.2 (10.8)	0.21
Medication Adherence, n (%)	High	63 (33.5%)	7 (20.6%)	56 (36.4%)	0.11
	Low/Medium	125 (66.5%)	27 (79.4%)	98 (63.6%)	

Abbreviations: IQR, interquartile range; PTSD, post-traumatic stress disorder; SD, standard deviation; OHSU, Oregon Health & Science University; VA, Veterans Affairs; CDAI, Clinical Disease Activity Index

tbl_summary() : table summary (1/2)

- What if I want one of those fancy summary tables that are at the top of most research articles?

```
1 library(gtsummary)
2 tbl_summary(dds.discr2)
```

Characteristic	N = 1,000 ¹
id	55,385 (31,759, 76,205)
age.cohort	
0-5	82 (8.2%)
6-12	175 (18%)
13-17	212 (21%)
18-21	199 (20%)
22-50	226 (23%)
51+	106 (11%)
age	18 (12, 26)
SAB	
Female	503 (50%)
Male	497 (50%)
expenditures	7,026 (2,898, 37,718)
R_E	
American Indian	4 (0.4%)
Asian	129 (13%)
Black	59 (5.9%)
Hispanic	376 (38%)
Multi Race	26 (2.6%)
Native Hawaiian	3 (0.3%)
Other	2 (0.2%)
White not Hispanic	401 (40%)
exp_to_age	462 (273, 938)
¹ Median (Q1, Q3); n (%)	

tbl_summary() : table summary (2/2)

- Let's make this more presentable

```
1 dds.discr2 %>%
2   select(-id, -age.cohort, -exp_to_age) %>%
3   tbl_summary(label = c(age ~ "Age",
4                         R_E ~ "Race/Ethnicity",
5                         SAB ~ "Sex Assigned at Birth",
6                         expenditures ~ "Expenditures",
7                         statistic = list(all_continuous() ~
8                                           "{mean} ({sd})"))
```

Characteristic	N = 1,000 [†]
Age	23 (18)
Sex Assigned at Birth	
Female	503 (50%)
Male	497 (50%)
Expenditures	18,066 (19,543)
Race/Ethnicity	
American Indian	4 (0.4%)
Asian	129 (13%)
Black	59 (5.9%)
Hispanic	376 (38%)
Multi Race	26 (2.6%)
Native Hawaiian	3 (0.3%)
Other	2 (0.2%)
White not Hispanic	401 (40%)
[†] Mean (SD); n (%)	

Resources

dplyr resources

- More `dplyr` functions to reference!

Additional details and examples are available in the vignettes:

- [column-wise operations vignette](#)
- [row-wise operations vignette](#)

and the dplyr 1.0.0 release blog posts:

- [working across columns](#)
- [working within rows](#)

R programming class at OHSU!

You can check out [Dr. Jessica Minnier's R class page](#) if you want more notes, videos, etc.

The larger tidy ecosystem

Just to name a few...

- janitor
- kableExtra
- patchwork
- gghighlight
- tidybayes

Credit to Mine Çetinkaya-Rundel

- These notes were built from Mine's notes
 - Most pages and code were left as she made them
 - I changed a few things to match our class
- Please see [her Github repository](#) for the original notes